

four predecessors and how long they walked earth

australopithecus afarensis: 3.9 million years ago

homo habilis: 2.4 million years ago

Homo erectus: 1.9 million years ago

homo neanderthalensis: 40,000 years ago

Ability that sets the human brain apart

symbolic thought + advanced language

Brain size of ancestor from 3 million years ago

400-450 cubic centimeters

Exhibit 2

a) Fish , Brain percent of body size: 0.1%

b)Reptile,brain percent of body size: 0.05%–0.1%

c)Human,brain percent of body size: 2%

d)dog, brain percent of body size: ~1%

e)Dolphin, brain percent of body size: ~1.3%

pink brain region in parts b–e

cerebrum

E better than b

Because the dolphin has a much larger & more folded cortex, it's better at: complex communication and problem solving

elephant brain weight calculation

Brain-to-body ratio = 1 : 150

Body weight = 900 kg

$$900 \div 150 = 6$$

#2

which major natural factor has the most impact on climate change?

volcanic eruptions

which major human factor has the most impact on climate change?

greenhouse gas emissions, co2 from fossil fuels

important takeaway from the global temperature animation

earth's temperature has risen a lot in the past century

yearly odds of an extreme heatwave in Europe

early 1900s: 1 in 1,000 chance per year

2015: about 1 in 10 chance per year

#3

object1: milky way galaxy central bulge, around 10,000 light years

obj2: Earth, 12,700 km diameter

obj3: Sun, 1,390,000 km in diameter

what happens when a star runs out of fuel

Its core collapses under its own gravity and if the star is massive enough, the collapse continues until a black hole forms

when was the first evidence of a black hole discovered, and what was its name

1971, cygnus x-1

how hot does an accretion disk get?

millions of degrees

what is gravitational lensing, and how does it relate to general relativity

gravitational lensing is the bending of light as it passes near a huge object, mass curves spacetime, light follows that curved path, creating effects like einstein rings